A PROJECT TITLE

ON

Customer Purchase Prediction Using Logistic Regression and Random Forest

project report submitted in partial fulfilment of the requirements for the Course

MACHINE LEARNING

BACHELOR OF TECHNOLOGY

in

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by

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DEPARTMENT OF Computer Science and Information Technology.

Name of Title: Customer Purchase Prediction Using Logistic Regression and Random Forest

1. Abstract:

Customer Purchase Prediction Using Logistic Regression and Random Forest is a machine learning project designed to predict whether a customer is likely to purchase a product based on their demographic information and purchasing behaviour. In today's competitive business environment, understanding customer buying patterns is essential for improving marketing strategies, increasing sales, and enhancing customer satisfaction. This project applies supervised machine learning techniques to analyse historical customer data and generate accurate purchase predictions that support informed business decisions.

The project begins with collecting and preprocessing customer data, which may include attributes such as age, gender, annual income, spending score, previous purchase history, and online activity. The dataset is cleaned by handling missing values, removing duplicate records, and encoding categorical variables. Exploratory Data Analysis (EDA) is then performed to identify trends, correlations, and important features influencing customer purchase decisions. The processed dataset is divided into training and testing sets to build and evaluate predictive models.

Two widely used classification algorithms, Logistic Regression and Random Forest, are implemented and compared. Logistic Regression provides a simple, interpretable statistical model that estimates the probability of a customer making a purchase, making it suitable for understanding feature relationships. Random Forest, an ensemble learning algorithm, constructs multiple decision trees and combines their predictions to improve accuracy, reduce overfitting, and capture complex patterns in customer behaviour. The performance of both models is evaluated using metrics such as accuracy, precision, recall, F1-score, and confusion matrix to determine the most effective approach.

The outcome of this project enables businesses to identify potential customers, optimise targeted marketing campaigns, improve customer engagement, and maximise revenue. By comparing the strengths and predictive capabilities of Logistic Regression and Random Forest, the project demonstrates how machine learning can transform raw customer data into actionable business insights. This solution provides a scalable and data-driven approach that supports intelligent decision-making in customer relationship management and modern retail analytics.

2.INTRODUCTION:

Customer purchase prediction is an important application of machine learning that helps businesses identify potential customers and improve marketing strategies. By analysing customer information such as age, gender, annual income, spending score, and purchase history, organisations can predict whether a customer is likely to purchase a product.

This project, **Customer Purchase Prediction Using Logistic Regression and Random Forest**, aims to develop a predictive model using supervised machine learning algorithms. The project involves data collection, preprocessing, exploratory data analysis (EDA), feature selection, model training, and performance evaluation.

Two classification algorithms are used in this project: **Logistic Regression** and **Random Forest**. Logistic Regression provides a simple and interpretable model for predicting purchase probability, while Random Forest improves prediction accuracy by combining multiple decision trees. The models are evaluated using performance metrics such as accuracy, precision, recall, F1-score, and confusion matrix.

The proposed system helps businesses understand customer behaviour, improve targeted marketing campaigns, and make data-driven decisions, ultimately increasing customer satisfaction and business profitability.

3.SOFTWARE REQUIREMENTS:

1) Operating System: Windows 10/11
2) Programming Language: Python 3.x
3) IDE: Jupyter Notebook / VS Code / Google Colab
4) Libraries: Pandas, NumPy, Matplotlib, Scikit-learn
5) Dataset: CSV File (.csv)

Signature of the Guide

Course Instructor